

APIA, Samoa

WMO: 917620

Lat: 13.815S

Lon: 171.781W

Elev: 7

StdP: 14.69

Time Zone: 13.00 (PCS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.5	72.8	65.4	93.9	76.0	67.0	99.2	76.3	25.3	82.5	23.5	82.1	2.4	180	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.3	88.2	79.8	87.6	79.7	87.1	79.4	81.3	86.5	80.8	86.3	80.4	85.9	11.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	154.1	85.3	79.1	151.0	84.9	78.7	148.8	84.6	45.0	87.1	44.4	86.6	44.0	86.2	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.2	21.0	18.9	DB	66.6	90.2	4.4	2.1	63.5	91.7	60.9	93.0	58.4	94.2	55.2	95.7
			WB	64.7	83.4	3.1	1.2	62.5	84.2	60.7	84.9	59.0	85.6	56.7	86.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	81.2	81.5	81.7	81.6	80.9	80.5	79.8	79.5	80.1	80.5	80.8	81.2
	DBStd	1.64	1.66	1.62	1.51	1.29	1.43	1.50	1.61	1.58	1.51	1.37	1.29	1.43
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11226	967	881	983	947	958	915	924	914	902	945	924	966
	CDD65	5751	502	461	518	497	493	465	459	449	452	480	474	501
	CDH74	58215	5225	4946	5588	5234	4969	4568	4362	4138	4385	4790	4806	5204
	CDH80	15706	1469	1459	1753	1607	1392	1204	994	903	1047	1207	1256	1415
(14) Wind	WSAvg	7.0	6.7	7.1	6.2	5.7	5.9	7.5	7.9	8.1	8.6	7.3	6.7	6.2
(15) Precipitation	PrecAvg	117.0	17.3	13.0	14.1	9.1	7.0	5.9	4.5	4.9	6.5	9.1	10.4	14.6
	PrecMax	160.4	37.8	36.8	27.4	20.8	14.4	12.9	11.5	12.1	23.3	22.8	28.6	36.7
	PrecMin	74.8	9.0	3.1	5.2	0.6	0.4	1.4	0.2	0.7	0.0	1.9	1.0	2.5
	PrecStd	25.1	7.5	6.1	5.7	4.9	3.2	3.1	2.6	3.1	5.0	6.0	6.1	8.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.6	89.1	88.9	89.0	87.9	87.3	86.4	86.2	86.6	87.2	87.3	87.7
		MCWB	79.7	80.1	80.3	80.2	79.6	78.9	77.9	77.9	78.2	78.5	78.9	79.4
	2%	DB	87.6	87.7	87.9	88.0	87.1	86.4	85.5	85.3	85.8	86.3	86.6	86.9
		MCWB	79.6	79.9	80.2	80.1	79.3	78.5	77.8	77.6	77.9	78.3	78.7	79.1
	5%	DB	86.6	86.9	87.2	87.2	86.4	85.8	85.0	84.7	85.2	85.6	86.0	86.2
		MCWB	79.3	79.6	79.9	79.7	79.0	78.2	77.5	77.2	77.6	78.0	78.5	78.9
	10%	DB	85.7	86.0	86.4	86.5	85.6	85.1	84.3	84.0	84.5	84.9	85.2	85.5
		MCWB	79.0	79.3	79.5	79.3	78.7	77.9	77.2	76.8	77.2	77.7	78.2	78.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	81.7	81.9	81.8	81.1	80.6	80.0	79.8	79.9	80.1	80.4	81.1
		MCDB	86.4	87.1	86.9	87.8	86.0	85.9	84.8	84.4	85.5	85.4	85.8	85.7
	2%	WB	80.8	80.9	81.1	80.9	80.3	79.8	79.1	78.9	79.1	79.3	79.8	80.3
		MCDB	86.1	86.3	86.5	86.9	85.8	85.2	84.4	84.1	84.7	84.9	85.3	85.4
	5%	WB	80.1	80.2	80.6	80.3	79.8	79.1	78.5	78.2	78.4	78.9	79.2	79.7
		MCDB	85.3	85.5	86.2	86.2	85.2	84.6	83.8	83.5	83.9	84.5	84.7	84.8
	10%	WB	79.5	79.7	79.9	79.8	79.2	78.5	77.9	77.5	77.9	78.3	78.7	79.1
		MCDB	84.5	84.9	85.4	85.4	84.5	84.0	83.1	82.8	83.3	83.8	84.2	84.3
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.6	9.3	9.7	9.4	9.3	9.1	9.1	8.9	8.8	8.7	8.4
		MCDBR	9.6	9.8	10.2	10.7	10.2	9.9	10.1	9.6	9.6	9.8	9.7	9.3
		MCWBR	4.7	4.9	5.2	5.5	5.2	5.2	5.3	5.4	5.3	5.5	5.0	4.8
	5% WB	MCDBR	8.9	9.1	9.7	10.1	9.7	9.2	9.1	9.1	8.9	9.2	9.0	8.7
		MCWBR	4.8	4.8	5.3	5.6	5.3	5.4	5.2	5.6	5.1	5.5	4.9	5.0
(40) Clear-Sky Solar Irradiance	taub		0.387	0.378	0.375	0.382	0.390	0.391	0.392	0.396	0.399	0.388	0.385	0.385
	taud		2.625	2.658	2.670	2.643	2.603	2.585	2.573	2.555	2.552	2.596	2.617	2.625
	Ebn at Noon		303	304	298	286	273	268	270	278	287	297	302	304
	Edh at Noon		32	31	30	29	29	28	29	31	33	33	32	32
(44) All-Sky Solar Radiation	RadAvg		1811	1810	1815	1632	1440	1376	1423	1649	1845	1903	1866	1842
	RadStd		181	169	135	133	101	70	57	78	104	96	98	115

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page